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Assignment No.09

1. Introduction –

The Send SMS Program allows users to send text messages from an app. Users can input a phone number and a message. Once they click "Send," the message is delivered via SMS. It provides a simple way to send texts easily.

2. Tools and Technologies used -

Tools and Technologies used in this Project are Android Studio, SDK compiler.

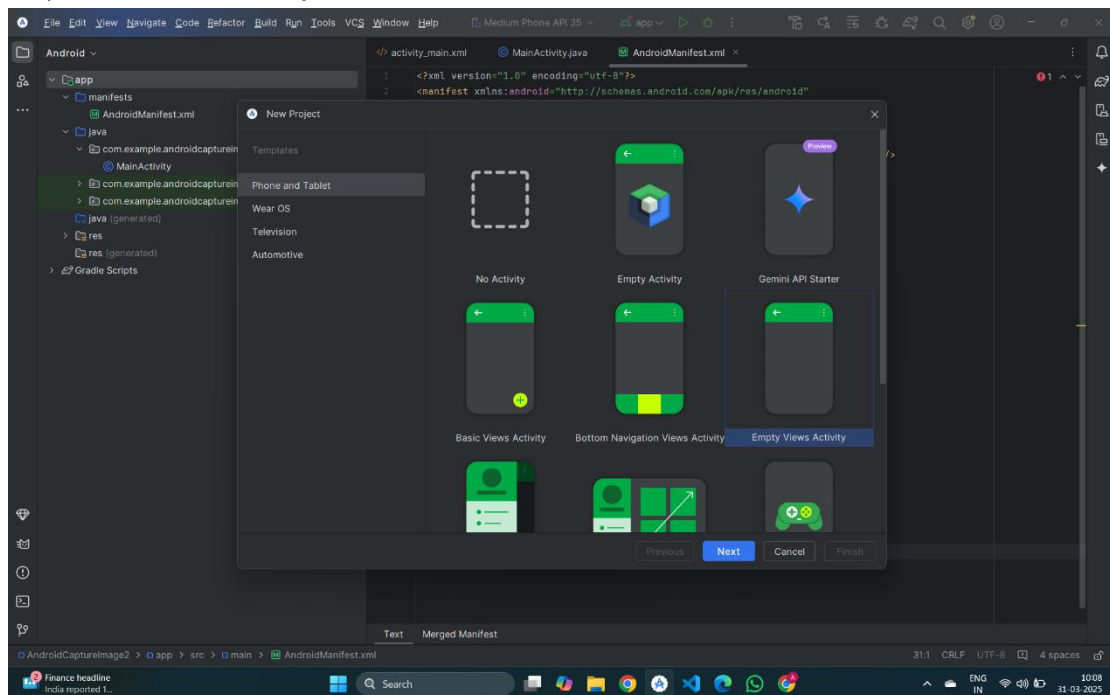
Mobile Emulator used - Medium phone API 35.

Language used – Java

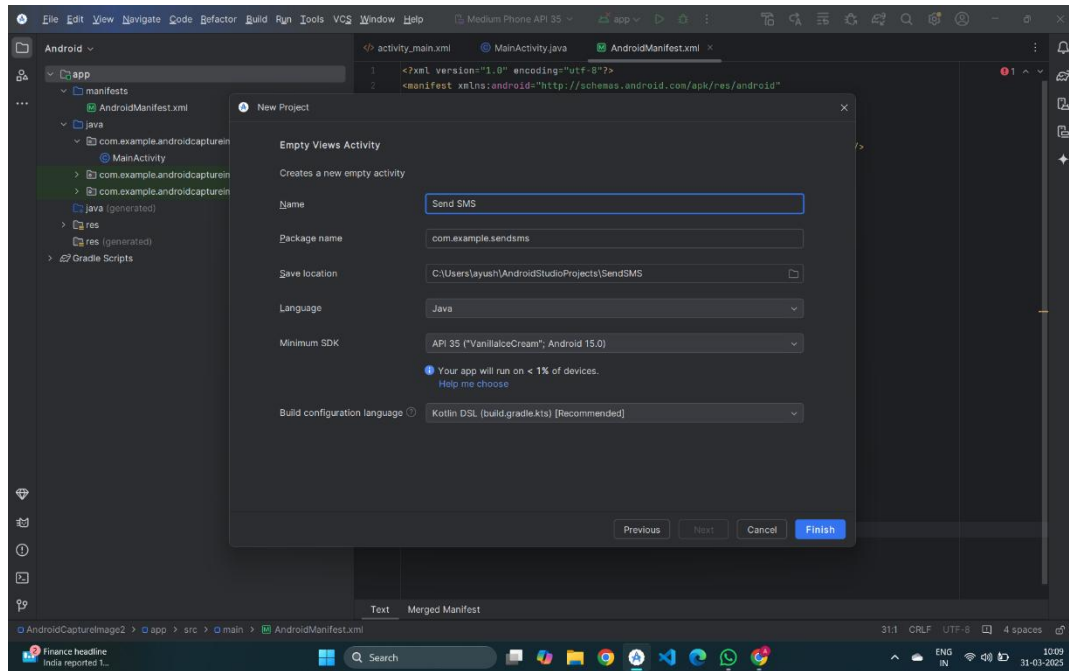
3. Procedure and steps –

The following steps were followed to complete the given task.

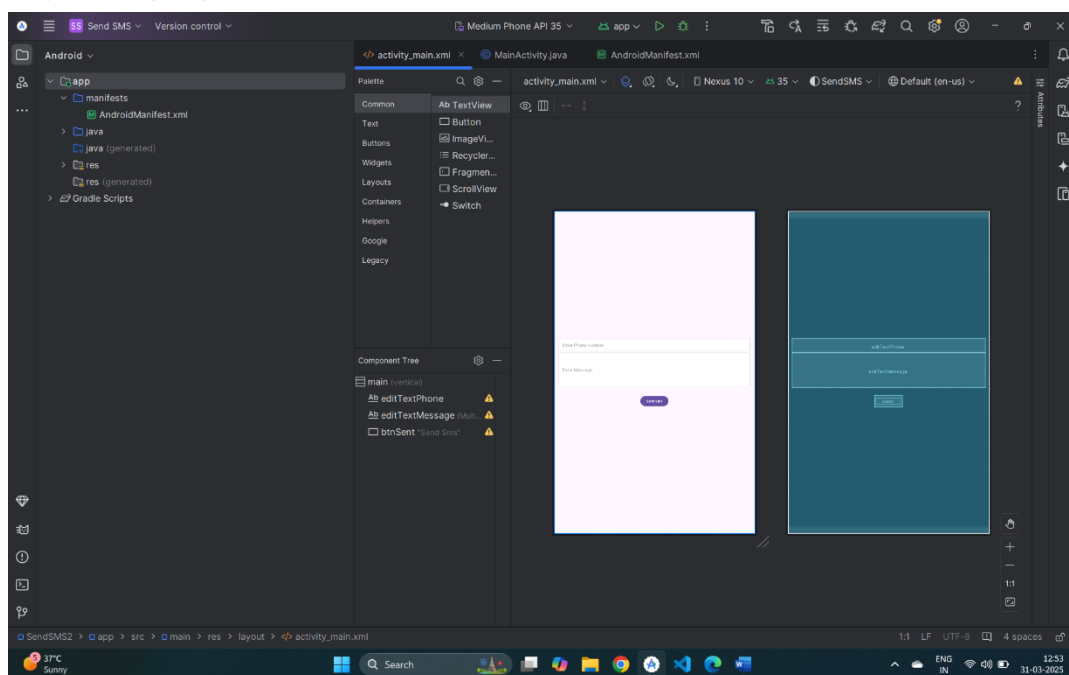
Step 1: Create a New Project:



Step 2: Naming and selecting language:



Step 3: Designing the UI :



Step 4: Writing the Code of xml:

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout
```

```
xmlns:android="http://schemas.android.com/apk/res/android"
```

```
android:id="@+id/main"
```

```
android:layout_width="match_parent"
```

```

        android:layout_height="match_parent"
        android:orientation="vertical"
        android:gravity="center"
        android:padding="16dp">

        <EditText
            android:id="@+id/editTextPhone"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Enter Phone number"
            android:padding="16dp"
            android:maxLength="10"
            android:background="@android:drawable/editbox_background"/>

        <EditText
            android:id="@+id/editTextMessage"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Enter Message"
            android:padding="16dp"
            android:inputType="textMultiLine"
            android:lines="5"
            android:background="@android:drawable/editbox_background"/>

        <Button
            android:id="@+id/btnSent"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Send Sms"
            android:textAllCaps="false"
            android:layout_marginTop="30dp"/>

    </LinearLayout>

```

Step 5: Writing the Code of AndroidManifest.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <uses-feature
        android:name="android.hardware.telephony"
        android:required="false" />

    <uses-permission android:name="android.permission.SEND_SMS"
        tools:ignore="PermissionImpliesUnsupportedChromeOsHardware" />

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"

```

```

        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.SendSMS"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>

```

Step 6: Writing code of mainactivity.java:

```

package com.example.sendsms;

import android.Manifest;
import android.content.pm.PackageManager;
import android.os.Bundle;
import android.telephony.SmsManager;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;

public class MainActivity extends AppCompatActivity {

    EditText editTextPhone, editTextMessage;
    Button btnSent;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}

```

```

        editTextPhone = findViewById(R.id.editTextPhone);
        editTextMessage = findViewById(R.id.editTextMessage);
        btnSent = findViewById(R.id.btnSent);

        btnSent.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                if (ContextCompat.checkSelfPermission(MainActivity.this,
Manifest.permission.SEND_SMS)
                    == PackageManager.PERMISSION_GRANTED) {
                    sendSMS();
                } else {
                    ActivityCompat.requestPermissions(MainActivity.this,
                        new String[]{Manifest.permission.SEND_SMS}, 100);
                }
            }
        });
    }

    @Override
    public void onRequestPermissionsResult(int requestCode, @NonNull String[] permissions,
@NonNull int[] grantResults) {
        super.onRequestPermissionsResult(requestCode, permissions, grantResults);
        if (requestCode == 100 && grantResults.length > 0 && grantResults[0] ==
PackageManager.PERMISSION_GRANTED) {
            sendSMS();
        } else {
            Toast.makeText(this, "Permission denied", Toast.LENGTH_SHORT).show();
        }
    }

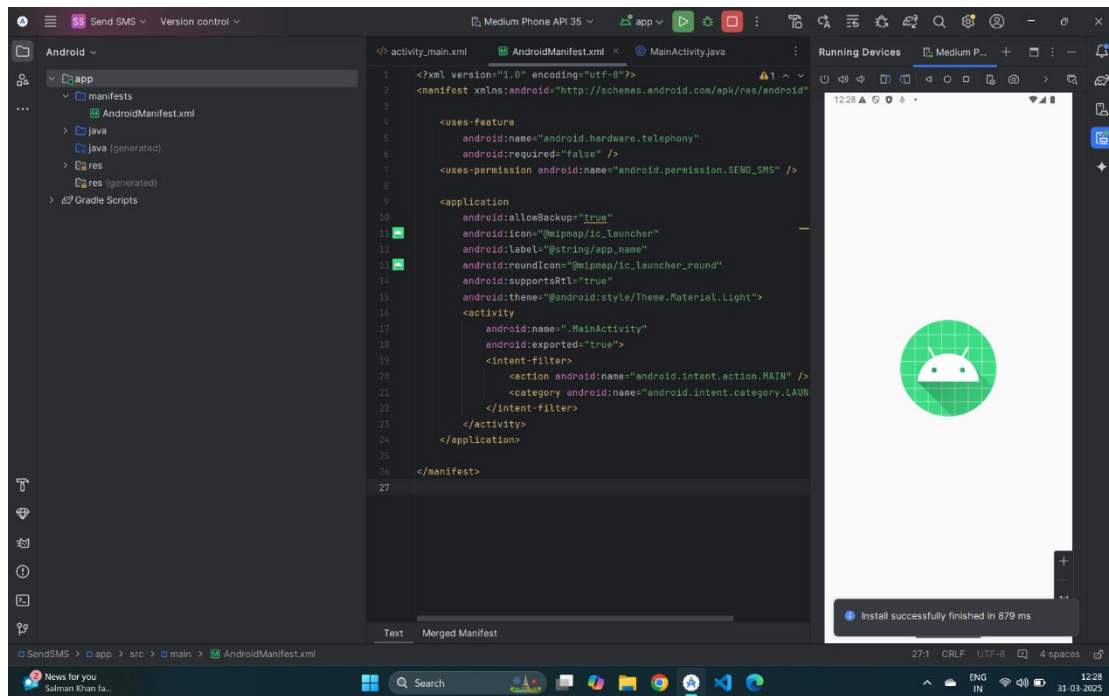
    private void sendSMS() {
        String phone = editTextPhone.getText().toString();
        String message = editTextMessage.getText().toString();

        if (!phone.isEmpty() && !message.isEmpty()) {
            try {
                SmsManager smsManager = SmsManager.getDefault();
                smsManager.sendTextMessage(phone, null, message, null, null);
                Toast.makeText(this, "SMS Sent", Toast.LENGTH_SHORT).show();
            } catch (Exception e) {
                Toast.makeText(this, "Failed to send SMS: " + e.getMessage(),
Toast.LENGTH_LONG).show();
            }
        } else {
            Toast.makeText(this, "Please enter phone number and message",
Toast.LENGTH_SHORT).show();
        }
    }

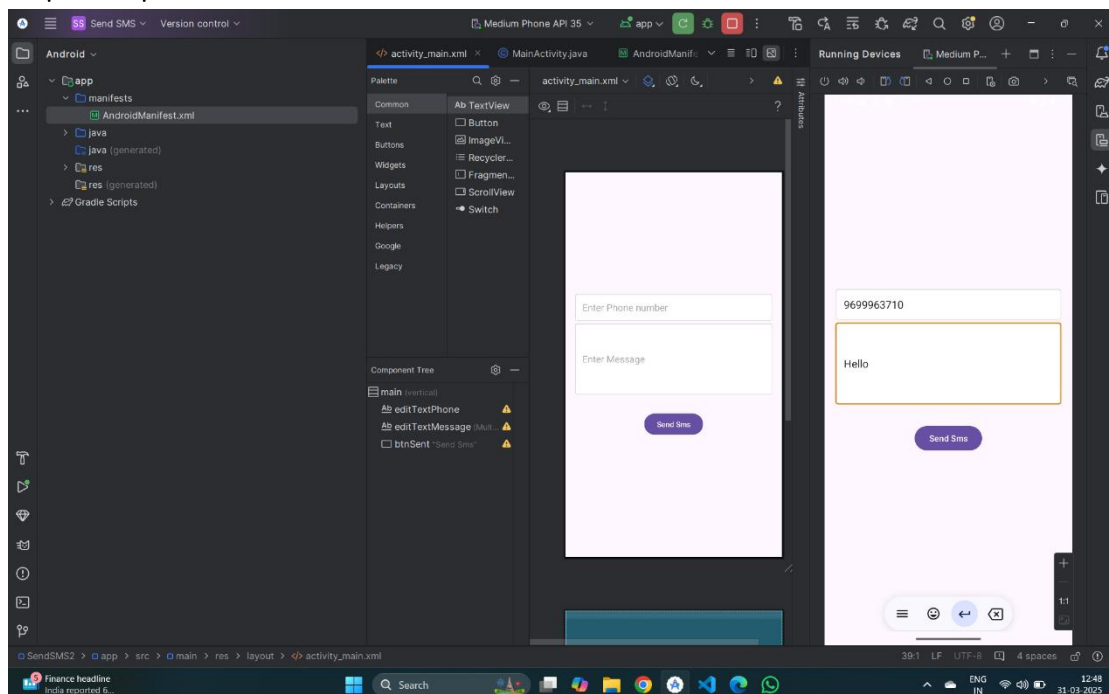
```

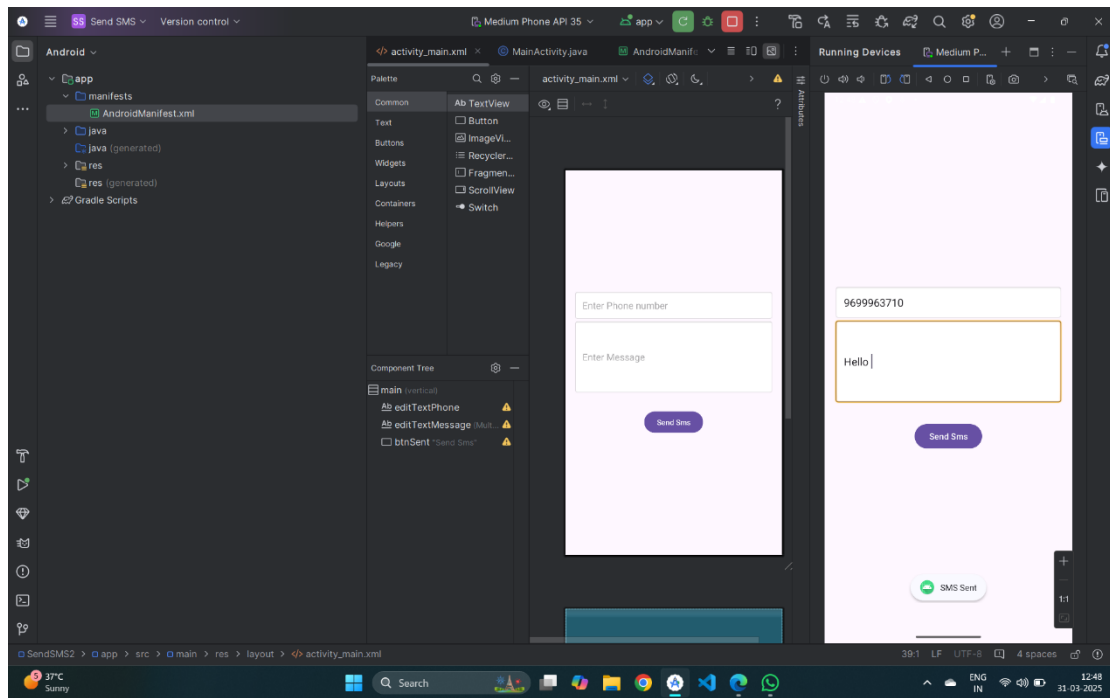
```
}  
}
```

Step 7: Running the application on Emulator



Step 8: Output:





Conclusion:

In conclusion, the Send SMS Program provides an easy and efficient way to send text messages directly from an application. With a user-friendly interface, it simplifies the process of messaging by allowing users to input phone numbers and messages quickly. This program is a great tool for automating communication and creating seamless messaging experiences.